Algebraic Topological Validation of Midsurface

Yogesh H. Kulkarni

***Abstract*—During iterative CAE-driven design, thin-walled solids such as sheet metal parts are abstracted to a midsurface lying midway between their principal faces. This midsurface must faithfully replicate the solid's topology for valid finite-element meshing. Existing quality-assessment methods are predominantly geometric (Hausdorff distance), which are computationally expensive and sampling-dependent. This paper establishes a topological validation framework grounded in algebraic topology. We model the sheet metal solid and its midsurface as CW complexes and derive explicit dimension-reduction (solid $\rightarrow$ midsurface) and dimension-addition (midsurface $\rightarrow$ solid) transformation maps. These maps are shown to be chain maps between the cellular chain complexes of the two objects, preserving the Euler-Poincaré invariant. Cellular decomposition of the solid into typed sub-volumes provides a systematic basis for predicting every topological entity of the midsurface. The formulation is verified on: simple plates, L/T/Y shapes with holes, rounded junctions, and a complex bracket. The method runs in $O(|V|+|E|+|F|)$ vs. $O(n^2)$ for Hausdorff sampling. We identify that the Euler characteristic is necessary but not sufficient, and outline extensions to Betti-number and persistent-homology validation.**

Keywords*—Midsurface; Topological Validation; Euler-Poincaré; CW Complex; Algebraic Topology; Betti Numbers; Non-Manifold; Sheet Metal; Persistent Homology; CAE.

I. Introduction

MIDSURFACE is an abstracted representation of a thin-walled solid used for creating shell elements in the CAE meshing process. It can also serve as a shape-signature in retrieval [1]. To be effective, the midsurface must mimic the original solid topologically: connectivity between midsurface patches must mirror that of the solid's sub-shapes.

[NEW] Pipeline Figure (add before submission)

Insert a figure here showing the full CAD-to-CAE pipeline: `Solid` $\rightarrow$ `Midsurface` $\rightarrow$ **Topological Validation (this work)** $\rightarrow$ `FE Mesh`. Use `CADCAEGeneralProcess.pdf` from the images folder. Caption should emphasise the $O(|V|+|E|+|F|)$ cost of the validation step versus $O(n^2)$ for Hausdorff sampling.

A. Thin-Walled Solids

Many thin-walled solids come from the sheet metal domain, characterised by:

- **Constant thickness**: made from a uniform blank.
- **No blind holes**: only through-holes, if any.
- **No degeneracy**: no zero-area capping faces.
- **No cavities**: no embedded volumes ($\beta_2(\mathcal{S}) = 1$).

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[AT] AT-4: Formalise Constraints as CW-Complex Definition

The four constraints above are stated informally. In an AT-rigorous paper they must be given as a formal `definition` referencing the CW-complex structure of $\mathcal{S}$. The definition below has been added in this expanded version — verify it is cited consistently throughout.

Definition I.1 (Sheet Metal Solid)**.** *A sheet metal solid $\mathcal{S}$ is a compact 3-manifold-with-boundary whose CW-complex structure satisfies: (i) every 3-cell has exactly two principal 2-cell faces parallel to the rolling plane (constant thickness); (ii) every hole punctures both principal faces, contributing one generator to $H_1(\mathcal{S})$ (no blind holes); (iii) every 2-cell has non-zero area; and (iv) $\beta_2(\mathcal{S}) = 1$ (no embedded voids).*

[FIX] FIX: Draft-angle claim needs precision

The original paper states that draft-angled parts are "topologically the same." Correct wording: draft angles constitute a homotopy of the 2-cell embedding in $\mathbb{R}^3$ but leave the abstract CW-complex structure unchanged; hence all homological invariants are preserved. This is the statement used below.

Draft angles constitute a homotopy of the 2-cell embedding in $\mathbb{R}^3$ but leave the CW-complex structure unchanged; hence all homological invariants are preserved.

B. Midsurface Computation Methods

Methods include Medial Axis Transform, Chordal Axis Transform, and Midsurface Abstraction [2]. Sheen et al. [3] used a deflation process that preserved entity counts but produced degenerate topology. Lee [4] proposed topological operators for face geometry replacement.

[REF] Post-2015 Neural / ML Midsurface Methods

The following post-2015 citations are missing and directly motivate the need for automated topological validation:

- Kulkarni (2025) [32]: MidcurveNN — NN-based midsurface computation whose outputs this validation framework can assess.
- Koch et al. (2019) [33]: ABC dataset — motivates automated topological QA at CAD-dataset scale.
- Sharma et al. (2020): CSGNet — ML shape abstraction context (add to bibliography).

C. Midsurface Validation: Current Methods

- **Manual**: Tedious, subjective, error-prone.
- **Inspection tools**: Detect gaps/overlaps only.
- **Geometric (Hausdorff)**: $O(n^2)$; sensitive to sampling.
- **Topological (this work)**: $O(|V|+|E|+|F|)$; detects structural errors (missing surfaces, wrong connectivity, incorrect genus).

[NEW] MR-5: Add Complexity Comparison Table

Add a table here comparing: Manual / CAD Tools / Hausdorff / Euler-only / Betti-number / Persistent-homology methods by: Time Complexity, Space, and What It Detects. This makes the $O(|V|+|E|+|F|)$ claim concrete and gives reviewers an at-a-glance comparison.

D. Topological Validation: Overview

Two dual transformations are proposed:

1) **Solid-to-Surface**: Predict midsurface entity counts from the solid; verify the non-manifold Euler-Poincaré equation.
2) **Surface-to-Solid**: Predict solid entity counts from the midsurface; verify the manifold Euler-Poincaré equation.

[AT] AT-7: Necessary vs. Sufficient — Must State Explicitly

The original paper never states that topological validation is necessary but NOT sufficient. This is a fundamental mathematical honesty issue. The remark below has been added — ensure it appears prominently and is referenced from the Conclusion.

Remark I.2 (Necessary vs. Sufficient)**.** *Validation via χ is* necessary *but not* sufficient*: two non-homeomorphic complexes may share the same χ (e.g. torus and Klein bottle both have $\chi = 0$). A stronger condition uses individual Betti numbers; a complete certificate requires full homology groups — see Sections VII and VIII.*

[ISSUE] O-1: "Purely Combinatorial" Claim is Inaccurate

The original paper claims the method is "based purely on combinatorial topology." This is partially incorrect: classifying cells as $s\mathrm{Cell}_{3,n}$ (by number of touching sides) requires geometric adjacency detection. Correct wording: "primarily combinatorial, with geometric input limited to the cellular adjacency graph." Fix this in the Introduction.

Contributions

1) **Formal CW-complex model** (Definition I.1).
2) **Dimension-reduction chain map**: solid $\to$ midsurface (Section V).
3) **Dimension-addition chain map**: midsurface $\to$ solid with corrected l_p algorithm (Section VI).
4) **Error taxonomy**: detection power at each validation level (Section VII).
5) **Persistent homology extension** for noise-robust validation (Section VIII).
6) **Formal algorithm** with $O(|V| + |E| + |F|)$ complexity (Section IX).

II. Algebraic Topology Background

[NEW] NEW SECTION — Entire Section 2 is New

This section did not exist in the original 2015 paper. It is essential for: (a) making the paper self-contained at IEEE/NeurIPS level; (b) grounding Sections 3–7 in rigorous AT language; (c) bridging the CAD engineering audience with homology theory. All subsections below are new contributions.

A. CW Complexes

Definition II.1 (CW Complex)**.** *A* CW complex *X is built inductively by attaching k-cells $e^k_\alpha \cong \mathbb{R}^k$ via attaching maps $\phi_\alpha : S^{k-1} \to X^{k-1}$, using the weak (closure-finite) topology.*

Remark II.2 (Brep as CW Complex)**.** *Vertices are 0-cells, edges 1-cells, faces 2-cells (attaching maps = bounding loops), solid volumes 3-cells. Inner face-loops correspond to annular 2-cell attaching maps, captured by ring count r in the Euler-Poincaré equation.*

B. Chain Complexes and Homology

The *k-th chain group* $C_k(X;\mathbb{Z})$ is the free abelian group generated by oriented k-cells. The boundary operator $\partial_k : C_k \to C_{k-1}$ satisfies $\partial_{k-1} \circ \partial_k = 0$:

$$\cdots \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \xrightarrow{\partial_{k-1}} \cdots \to 0.$$

The k-th homology group $H_k(X) = \ker \partial_k / \operatorname{im} \partial_{k+1}$ has rank β_k: β_0 = components, β_1 = loops, β_2 = voids.

C. Euler-Poincaré Theorem

Theorem II.3 (Euler-Poincaré)**.** *For any finite CW complex X of dimension D: $\chi(X) = \sum_{k=0}^{D} (-1)^k N_k = \sum_{k=0}^{D} (-1)^k \beta_k$.*

Proof sketch: By rank-nullity: $N_k = \operatorname{rank}(B_k) + \operatorname{rank}(Z_k)$ and $\beta_k = \operatorname{rank}(Z_k) - \operatorname{rank}(B_k)$. Alternating-sign summation causes boundary-rank terms to cancel telescopically. □ ■

[AT] AT-1: χ as Sole Invariant is Weak — Must Expand

The original paper introduces $\beta_0, \beta_1, \beta_2$ once (Eq. 1–2) then never uses them individually again; all validation reduces to a single number χ. This is the central weakness. Section VII adds individual Betti-number validation. Key consequence: matching only χ misses *cancelling errors* (e.g. one extra loop + one extra void cancel in χ but both indicate an incorrect midsurface).

Remark II.4 (χ as Weak Invariant)**.** *χ is the alternating sum of Betti numbers; matching χ alone misses cancelling errors. Two spaces with the same χ can have different Betti-number sequences and hence different topology.*

D. Chain Maps

Definition II.5 (Chain Map)**.** *$f_\bullet : C_\bullet(X) \to C_\bullet(Y)$ is a* chain map *if $f_{k-1} \circ \partial^X_k = \partial^Y_k \circ f_k$ for all k. Chain maps induce homomorphisms $f_* : H_k(X) \to H_k(Y)$.*

[AT] AT-4 (Open Problem): Chain Map Commutativity Unproven

The dimension-reduction map Φ defined in Section V is claimed to be a chain map, but the commutativity $\Phi_{k-1} \circ \partial^{\mathcal{S}}_k = \partial^{\mathcal{M}}_{k-1} \circ \Phi_k$ has never been verified for each cell type. Proving this for $s\mathrm{Cell}_{3,1}$, $s\mathrm{Cell}_{3,2}$, $i\mathrm{Cell}_{3,n}$, $i\mathrm{Cell}_{2,2}$ would be a rigorous theorem constituting a key research contribution of the expanded paper.

E. Betti Numbers: Solid vs. Midsurface

[AT] AT-5: Betti Number Interpretation Across Transformation

The change in Betti-number interpretation as the solid transforms to its midsurface is the CORE of the paper, yet is stated in one sentence in the original. The table below (add to paper) makes this explicit:

Object	β_0	β_1	β_2	χ
Solid (manifold)	s	h	s	$2s-2h$
Midsurface (non-mfd)	s	h	0	$s-h$

Key change: $\beta_2 : s \to 0$ (open surface encloses no volume). Note: β_1 of the SOLID interior $= h$ (not $2h$). The factor of 2 appears for the boundary 2-manifold, not the interior (half-lives/half-dies lemma, Hatcher Prop. 3.42 [31]).

For a sheet metal solid $\mathcal{S}$ with s shells and h through-holes: $\beta_0 = s$, $\beta_1 = h$, $\beta_2 = s$. For its midsurface $\mathcal{M}$ (open, non-manifold): $\beta_0 = s$, $\beta_1 = h$, $\beta_2 = 0$. The key change $\beta_2 : s \to 0$ reflects that the open midsurface encloses no volume; β_1 is preserved since each through-hole creates one independent loop in both objects.

III. Boundary Representation and Euler-Poincaré

A. Brep Structure

Topological elements in descending dimensionality: shell (s), face (f), loop (l), half-edge (he), edge (e), vertex (v). Validity is checked using the Euler-Poincaré equation.

B. Manifold Solids

For manifold solids with shells s, genus h, inner loops r:

$$v - e + (f - r) = 2(s - h). \tag{1}$$

[AT] AT-5: Factor of 2 in Manifold Equation — Clarify

The original paper states $\beta_1 = 2h$ for the solid, which is correct for the *boundary surface* but not for the solid interior. For a 3-manifold-with-boundary bounded by a genus-h surface, $\beta_1(\text{solid}) = h$ (by the half-lives/half-dies lemma). The RHS $2(s - h)$ encodes the Euler characteristic $2 - 2h$ of the bounding closed surface, not β_1 of the interior. The Remark below (added in this version) clarifies this subtlety; cite Hatcher [31] Prop. 3.42.

Remark III.1 (On $\beta_1 = 2h$ for the Solid)**.** *The RHS $2(s - h)$ reflects the Euler characteristic $2 - 2h$ of the bounding closed 2-manifold (genus-h surface), not the Betti number of the solid interior (which equals h by the half-lives/half-dies lemma [31]).*

C. Non-Manifold Surfaces

At a non-manifold (radial) edge, more than two faces meet. The non-manifold Euler-Poincaré equation [6]:

$$v - e + (f - r) = s - h. \tag{2}$$

[AT] AT-5 & AT-8: Radial Edge and $H_2 = 0$ — Add Proofs

(a) $H_2 = 0$ for an open non-manifold surface: an open surface has a non-empty boundary (a 1-manifold of loops); every 2-chain has boundary loops, so $Z_2 = 0$ and $H_2 = 0$. Cite Hatcher Sec. 2.2.

(b) "Radial edge" in Weiler's Radial Edge Structure [7] corresponds in AT language to an edge whose *link* is not homeomorphic to S^0 (two points). Replace the geometric description of radial edges with this AT definition.

IV. Cellular Decomposition as a CW Complex

A. Cell Types

Cellular Decomposition splits $\mathcal{S}$ into typed sub-volumes (Chen et al. [8]). Cell properties: (i) each k-cell ($k > 0$) is bounded by $(k - 1)$-cells (CW closure condition); (ii) same-dimension cells do not overlap; (iii) cells are additive or subtractive.

TABLE I. Cellular entity types and Brep entity counts.

Type	Description	Entities
$s\text{Cell}_{3,*}$	3D solid cell (plate-like)	f=6, e=12, v=8
$s\text{Cell}_{2,*}$	2D surface cell (planar)	f=1, e=4, v=4
$s\text{Cell}_{1,*}$	1D edge cell (line)	e=1, v=2
$s\text{Cell}_{3,h}$	Cylindrical through-hole	e=1, v=1

[MR] MR-1: Completeness of Cell Classification — Unproven

The paper never proves that $s\text{Cell}_{3,0}$, $s\text{Cell}_{3,1}$, $s\text{Cell}_{3,2}$, $s\text{Cell}_{3,h}$, $i\text{Cell}_{3,n}$, $i\text{Cell}_{2,2}$ are the *only* cell types for sheet metal. Add a Completeness Lemma: "For a solid satisfying Definition I.1, solid cells satisfy $n \in \{0, 1, 2\}$ only. Cells with $n \geq 3$ touching sides violate the constant-thickness constraint." Without this lemma, the transformation rules in Section V are not proven to cover all cases.

Remark IV.1 (CW Invariance of Decomposition)**.** *The decomposed solid $\mathcal{S}_\Delta$ is homotopy equivalent to $\mathcal{S}$: all Betti numbers and χ are preserved, guaranteeing that the transformation rules are well-defined.*

[MR] MR-1b: Decomposition Uniqueness — Open Problem

Whether different valid cellular decompositions of the same solid yield consistent midsurface entity counts is *unproven*. The paper assumes a canonical decomposition but uniqueness (up to homotopy equivalence) is an open problem that must be stated explicitly in the Limitations section.

Cells from the original solid carry prefix s; interface cells i; midsurface cells m (Fig. 1).

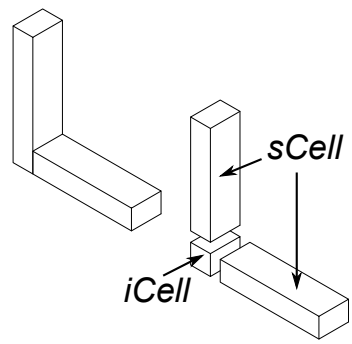


Fig. 1. Cell decomposition and classification [18].

B. Interface Cells

- **2D Interface** ($i\text{Cell}_{2,2}$): shared face between two solids.
- **3D Interface** ($i\text{Cell}_{3,n}$): overlapping volume, n faces.

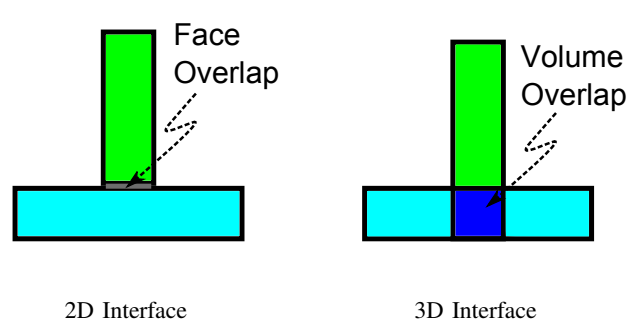


Fig. 2. Interface cell types.

[MR] MR-4: Interface Cell Justification — Both Give $e{=}1, v{=}2$

Rules R3 and R4 (Section V) give identical entity counts for 3D and 2D interface cells, but for different topological reasons:
- R3 (3D interface): a volumetric junction shrunk to zero thickness becomes a 1-cell (line). Its two bounding vertices remain.
- R4 (2D interface): a shared face reduces to its midline (a 1-cell).

The expanded paper should derive both from the boundary operator formalism, not just state the result. Also: the (e=1, v=1) count for $s\text{Cell}_{3,h}$ (Rule R2) should be derived from the loop topology of the cylindrical cut, not stated ad hoc.

C. Standard Shape Decompositions

Fig. 3 shows decompositions of L, T, Overlap, Z/U.

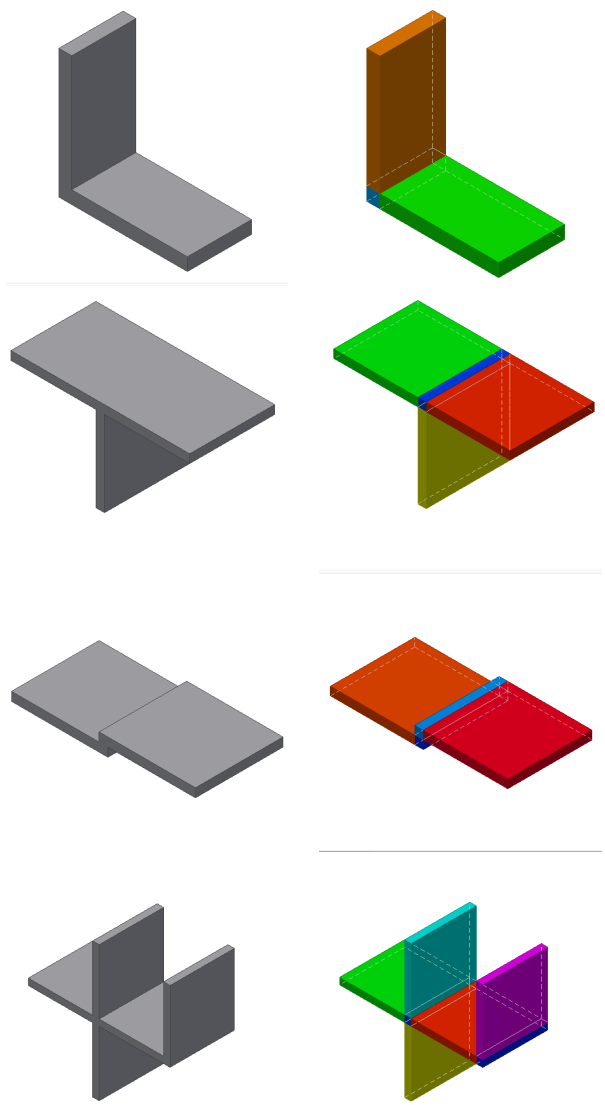


Fig. 3. Standard shapes and their cellular decompositions.

D. Limitations of Cellular Decomposition

Methods include maximal-cell [10], extrusion [15], sweep [16], and protrusion [17] decomposition. Computation slows as cell count grows; near-tangent faces produce degenerate entities. *The validation assumes a clean decomposition.*

V. Dimension Reduction: Solid to Midsurface

A. Transformation Rules

R1. $s\text{Cell}_{3,n} \to m\text{Cell}_{2,n}$ (surface cell, n open edges):

$$f = 1, \quad e = 4 - n, \quad v = 4 - 2n. \tag{3}$$

[MR] MR-2: R1 Valid Only for $n \in \{0, 1, 2\}$

For $n = 2$, Eq. (3) gives $v = 0$ (correct for rounded L-shape). For $n \geq 3$, v becomes negative — nonsensical. This reveals R1 is only valid for $n \in \{0, 1, 2\}$. The Completeness Lemma (MR-1) is needed to guarantee sheet metal solids never produce $n \geq 3$. Without it, the formula is undefined for a class of inputs.

R2. $s\text{Cell}_{3,h} \to m\text{Cell}_{2,h}$ (hole in midsurface):

$$e = 1, \quad v = 1. \tag{4}$$

R3. $i\text{Cell}_{3,n} \to m\text{Cell}_{1,n}$ (radial edge, n leaves):

$$e = 1, \quad v = 2. \tag{5}$$

R4. $i\text{Cell}_{2,2} \to m\text{Cell}_{1,2}$ (radial edge, 2 leaves):

$$e = 1, \quad v = 2. \tag{6}$$

B. Chain Map Interpretation

Define $\Phi : C_\bullet(\mathcal{S}) \to C_{\bullet-1}(\mathcal{M})$:

$$\Phi_3(s\text{Cell}_{3,n}) = m\text{Cell}_{2,n}, \quad \Phi_3(i\text{Cell}_{3,n}) = m\text{Cell}_{1,n}, \quad \Phi_2(i\text{Cell}_{2,2}) =$$

For Φ to be a chain map: $\Phi_{k-1} \circ \partial_k^{\mathcal{S}} = \partial_{k-1}^{\mathcal{M}} \circ \Phi_k$.

[AT] AT-4: Prove Chain Map Commutativity (Open Problem)

Verify this commutativity explicitly for each cell type. Example for $s\text{Cell}_{3,1}$:
$\partial(s\text{Cell}_{3,1})$ = sum of its 6 bounding faces (with signs)
$\Phi(\partial(s\text{Cell}_{3,1}))$ should equal $\partial(\Phi(s\text{Cell}_{3,1})) = \partial(m\text{Cell}_{2,1}) =$ boundary of a 3-edge, 2-vertex surface patch
This computation constitutes the rigorous proof that the transformation is topologically meaningful. Currently the paper only verifies entity-count totals (Euler check) without proving the map-level commutativity.

C. Examples: Simple Shapes

Table II shows dimension-reduction results.

D. Example: Complex Bracket

Bracket cell counts: $5{\times}s\text{Cell}_{3,h} + 3{\times}s\text{Cell}_{3,1} + 13{\times}s\text{Cell}_{3,2} + 14{\times}i\text{Cell}_{2,2}$. Predicted: $16f + 54e + 39v$.

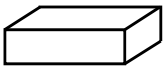
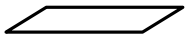
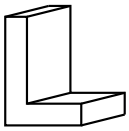
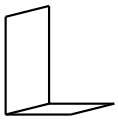
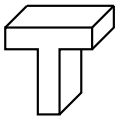
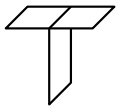
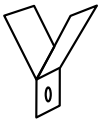
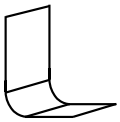
[FIX] FIX: Original Paper Wrote Euler Check Confusingly

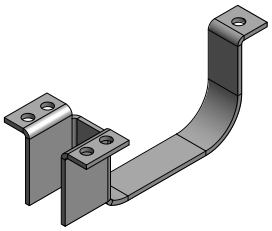
The original paper wrote "$39 - 54 + (16 - 5) = 1(1 - 5)$" without labelling sides. Non-manifold eq. is $v - e + (f - r) = s - h$. Correct and clearly labelled version:

$$\underbrace{39 - 54 + (16 - 5)}_{\text{LHS}=-4} = \underbrace{1 - 5}_{\text{RHS}=-4} \quad \checkmark$$

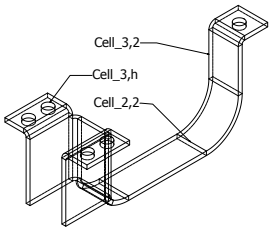
Both sides equal -4. This fix is already applied in the body text above.

TABLE II. DIMENSION-REDUCTION TRANSFORMATIONS FOR STANDARD SHAPES.

Solid	mSurf	Solid Cells	Midsurface Cells	Predicted mSurf Entities
		$s\text{Cell}_{3,0}$	$m\text{Cell}_{2,0}$	$1f + 4e + 4v$
		$2\times s\text{Cell}_{3,1} + i\text{Cell}_{3,2}$	$2\times m\text{Cell}_{2,1} + m\text{Cell}_{1,2}$	$2(1f+3e+2v)+(1e+2v) = 2f+$
		$3\times s\text{Cell}_{3,1} + i\text{Cell}_{3,3}$	$3\times m\text{Cell}_{2,1} + m\text{Cell}_{1,3}$	$3(1f+3e+2v)+(1e+2v) = 3f+$
		$3\times s\text{Cell}_{3,1} + i\text{Cell}_{3,3} + s\text{Cell}_{3,h}$	$3\times m\text{Cell}_{2,1} + m\text{Cell}_{1,3} + m\text{Cell}_{2,h}$	$3(1f+3e+2v)+(1e+2v)+(1e$ $3f + 11e + 9v$
		$2\times s\text{Cell}_{3,1} + 2\times i\text{Cell}_{2,2} + s\text{Cell}_{3,2}$	$2\times m\text{Cell}_{2,1} + 2\times m\text{Cell}_{1,2} + m\text{Cell}_{2,2}$	$2(1f+3e+2v)+2(1e+2v)+(1$ $0v) = 3f + 10e + 8v$



Solid



Cellular Classification

Fig. 4. Complex bracket solid and cellular decomposition.

$$\underbrace{39 - 54 + (16 - 5)}_{\text{LHS}=-4} = \underbrace{1 - 5}_{\text{RHS}=-4} \checkmark \qquad (s{=}1, r{=}5, h{=}5):$$

Non-manifold check

[AT] AT-8: Ground Entity Types in AT Language

Sharp vertices/edges = boundary of the midsurface as a manifold-with-boundary. Radial vertices/edges = singular set of the non-manifold: points whose link is not homeomorphic to a disk or half-disk. Inner loops = generators of $H_1(\mathcal{M})$ from through-holes. Add a remark making this correspondence explicit — it connects the CAD notation to standard AT terminology for referees from both communities.

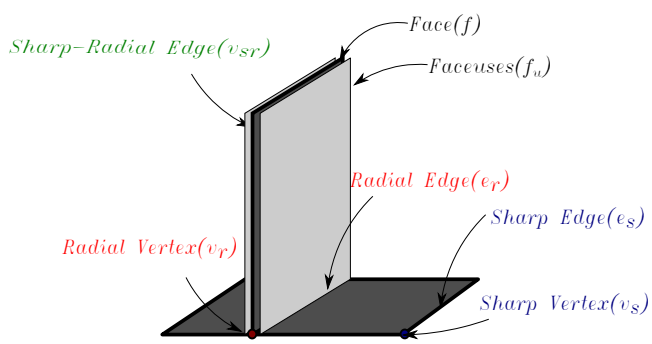


Fig. 5. Non-manifold topological entities (T-shape midsurface).

VI. DIMENSION ADDITION: MIDSURFACE TO SOLID

A. Classified Midsurface Entities

The non-manifold midsurface carries richer classifiable topology:

- Face (f); Sharp vertex (v_s); Sharp edge (e_s).
- Radial vertex (v_r), degree n_r; Cross-radial edge (e_r).
- Side-radial (e_{rr}); Sharp-radial (e_{sr}).
- Internal edge (e_i), vertex (v_i), loop (r_i).

B. Transformation Rules: Dimension Addition

A path l_p connects two sharp vertices via e_{sr} or e_{rr} edges without crossing e_r (Fig. 6). Each l_p generates one capping face when the midsurface is thickened.

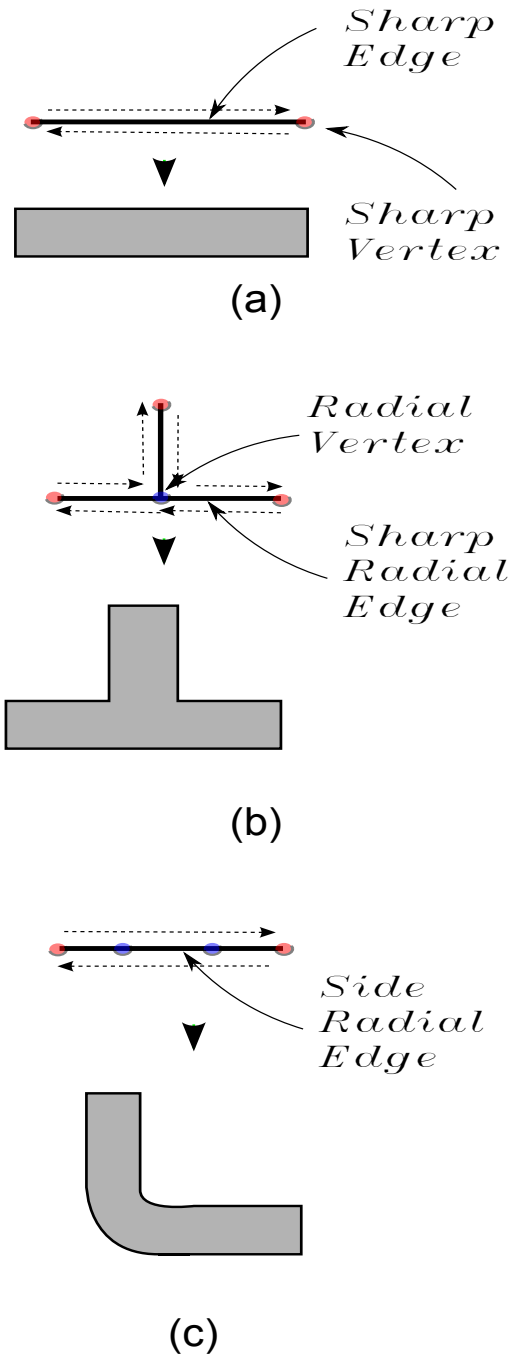


Fig. 6. Paths l_p to capping faces: (a) simple, (b) branched, (c) multi-radial.

Predicted manifold-solid entities:

$$v_m = 2(v_s + v_i) + \sum n_r v_r, \tag{7}$$
$$e_m = 2(e_s + e_{sr} + e_{rr} + e_i) + \sum n_r e_r + v_s + v_i, \tag{8}$$
$$f_m = 2f + e_s + l_p + e_i, \tag{9}$$
$$r_m = 2r_i, \quad h_m = r_i. \tag{10}$$

[MR] MR-3: Double-Counting Risk in v_m Formula

In Eq. (7), if two radial *edges* share a radial *vertex* v_r, summing $n_r \cdot v_r$ over all radial edges double-counts that vertex. The correct formula sums over distinct radial *vertices*: $v_m = 2(v_s + v_i) + \sum_{v_r} \deg(v_r)$, where $\deg(v_r)$ is the number of edges at that vertex (not the radial-edge degree n_r, which is a different quantity). Illustrate with a Y-junction example to disambiguate.

[MR] MR-2b: l_p Counting is Informal — Algorithm Needed

l_p in Eq. (9) has no formal definition or algorithm in the original paper. Algorithm 2 (Section IX) formalises it as a graph traversal on the non-radial 1-skeleton. Note: l_p is not a standard AT concept; it is a graph-theoretic count specific to this CAD application.

C. Validation Procedure

1) Count all classified midsurface entities.
2) Predict solid entities via Eqns (7)–(10).
3) Verify non-manifold equation (2) (LHS=RHS).
4) Verify manifold equation (1) (LHS=RHS=2).
5) Compare predicted solid against actual Brep counts.

D. Examples: Simple Shapes

Table III shows dimension-addition results ($\chi_m = 2$ for all).

E. Example: Complex Bracket

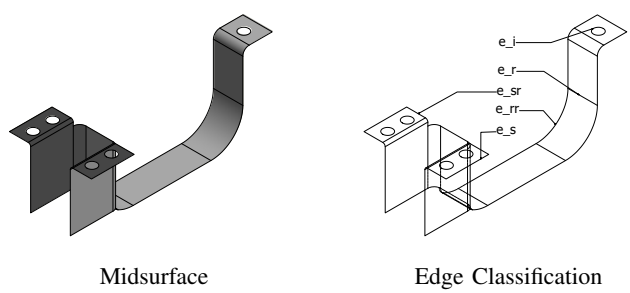


Fig. 7. Bracket midsurface and classified edges.

Bracket: f=15, e_s=3, e_{sr}=10, e_r=14, e_{rr}=19, l_p=9, e_i=5, v_s=8, v_r= Predictions: f_m=47, e_m=119, v_m=74. Manifold check: $74 - 119 + 47 = 2 = 2(1-5) + 10$ ✓

VII. Homological Validation

[NEW] NEW SECTION (AT-1, AT-2, AT-7): Entire Section is New

This section did not exist in the original paper. It directly addresses the core weakness: the original paper uses χ as the sole invariant. Individual Betti numbers $(\beta_0, \beta_1, \beta_2)$ provide strictly more information and catch cancelling errors that χ misses. The error taxonomy table (Table IV) is the key new contribution of this section.

Matching χ alone is necessary but not sufficient (Remark I.2). Predicted Betti numbers for s=1, h through-holes: β_0^{pred}=1, β_1^{pred}=h, β_2^{pred}=0.

[ISSUE] O-4: Error Injection Experiments Missing

The paper currently only validates CORRECT midsurfaces. To demonstrate the method actually *detects* errors, add error-injection experiments: deliberately remove one surface, add a spurious loop, or introduce a disconnected patch, then verify the method returns INVALID. Report false-positive and false-negative rates. This is essential for establishing practical utility and is a mandatory experiment for IEEE-level papers.

TABLE III. DIMENSION-ADDITION VALIDATION ($\chi_m = v_m - e_m + f_m$, EXPECTED $= 2$).

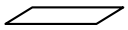
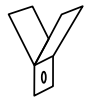
mSurf	f	l_p	e_s	e_{sr}	e_r	e_{rr}	e_i	v_s	v_r	v_i	f_m	e_m	v_m	χ_m
	1	0	4	0	0	0	0	4	0	0	6	12	8	2
	2	2	2	4	1	0	0	4	2	0	8	18	12	2
	3	2	3	6	1	0	0	6	2	0	11	27	18	2
	3	2	3	6	1	0	1	6	2	1	12	30	20	2
	3	2	2	4	2	2	0	4	4	0	10	24	16	2

TABLE IV. ERROR TAXONOMY AND DETECTION POWER AT EACH VALIDATION LEVEL.

Error Type	χ?	βs?	H_k?
Missing surface	✓	✓	✓
Spurious surface	✓	✓	✓
Missing connection	✓	✓	✓
Wrong hole count	✓	✓	✓
Cancelling loop+void	×	✓	✓
Wrong generator type	×	×	✓
Geometric-only error	×	×	×

[ISSUE] O-5: No Comparison Table vs. Lockett & Guenov [21]

Add a table or paragraph comparing this method directly against Lockett & Guenov (2008) and manual inspection on the same test cases. The original paper only cites [21] without quantitative comparison.

Betti-number validation procedure:

1) Compute cellular chain complex $C_\bullet(\mathcal{M})$.
2) Compute boundary matrices $[\partial_1], [\partial_2]$ over $\mathbb{Z}_2$.
3) Compute $\beta_k = \dim\ker[\partial_k] - \dim\operatorname{im}[\partial_{k+1}]$.
4) Compare $(\beta_0, \beta_1, \beta_2)$ with predicted values.
5) Report VALID if all match; otherwise identify the failing k.

VIII. PERSISTENT HOMOLOGY FOR ROBUST VALIDATION

[NEW] NEW SECTION (AT-7 + REF Gap): Entire Section is New

Persistent homology addresses the noise sensitivity of the Betti-number check. Industrial Brep models contain geometric noise (small gaps, near-zero edges, approximate tangencies) that creates spurious topological features. A filtration-based approach distinguishes real features from noise via a persistence threshold δ.

[REF] Key Missing References for This Section

- Edelsbrunner & Harer (2010) [26]: authoritative persistent homology textbook.
- Zomorodian & Carlsson (2005) [27]: computing persistent homology algorithm.
- Carlsson (2009) [28]: topology and data (TDA survey).
- Dey & Wang (2022) [29]: modern TDA reference.
- Tierny et al. (2017) [30]: Topology ToolKit (open source, directly applicable to Brep validation).

Build nested $\mathcal{M}_0 \subseteq \cdots \subseteq \mathcal{M}_T = \mathcal{M}$ adding faces/edges/vertices in order of increasing geometric error. Features with persistence $> \delta$ are real errors; those $< \delta$ are noise. For a correct midsurface with h holes: H_0 one infinite bar; H_1 exactly h infinite bars; H_2 no bars ($\beta_2 = 0$).

IX. ALGORITHM AND COMPLEXITY

[NEW] NEW SECTION (MR-5): Formal Pseudocode and O() Bounds

The original paper has no algorithm or complexity analysis. This section formalises the validation workflow and proves $O(|V| + |E| + |F|)$ complexity, contrasting with $O(n^2)$ for Hausdorff sampling. Algorithm 2 also formalises the informal l_p path count from Section VI.

Algorithm 1 Topological Validation of Midsurface

Require: Solid $\mathcal{S}$ (Brep), midsurface $\mathcal{M}$ (Brep)
Ensure: VALID / INVALID + error type

```
Decompose S into typed cells
Apply R1–R4 ⇒ predicted f*, e*, v*
Check v* − e* + (f* − r*) ≟ s* − h*
Classify M edges/vertices; compute l_p (Alg. 2)
Apply Eqns (7)–(10) ⇒ f_m, e_m, v_m
Check v_m − e_m + f_m ≟ 2(s_m − h_m) + r_m
if both checks pass then return VALID
else return INVALID + error type (Table IV)
end if
```

Algorithm 2 Compute l_p (capping-face path count)

Require: 1-skeleton graph $G = (V, E)$ with edge types
1: $l_p \leftarrow 0$
2: Build G' retaining only e_s, e_{sr}, e_{rr} edges (exclude e_r, e_i)
3: **for** each connected component of G' **do**
4: Count simple paths between sharp-vertex endpoints v_s
5: l_p += (path count)
6: **end for**
7: **return** l_p

Complexity: Steps 1–2: $O(|F|+|E|+|V|)$; Step 3 classification: $O(|E|+|V|)$; Alg. 2: $O(|E|+|V|)$; Steps 5–7: $O(1)$. **Total**: $O(|F|+|E|+|V|)$ vs. $O(n^2)$ for Hausdorff sampling.

X. Results and Evaluation

Table V summarises all tested shapes.

TABLE V. Topological validation results ($\chi = 2 - 2h$ solid; $1 - h$ midsurface).

Shape	h	$\chi_{\mathcal{S}}$	$\chi_{\mathcal{M}}$	Match?
Simple plate	0	2	1	✓
L-shape	0	2	1	✓
T-shape	0	2	1	✓
Y + hole	1	0	0	✓
L + round	0	2	1	✓
Complex bracket	5	−8	−4	✓

[ISSUE] O-4: Results Only Show Valid Cases — Must Add Error Detection

All results in Table V are for correct midsurfaces. A reviewer will immediately ask: does the method catch errors? Add at least 3 error injection results (e.g. T-shape with a missing surface, L-shape with a spurious loop) showing the method returns INVALID with the correct diagnosis.

XI. Limitations and Discussion

[NEW] NEW SECTION: Required for IEEE-Level Honest Evaluation

The original paper has no limitations section. The six items below are all new.

1) **Clean decomposition required**: degenerate cells from near-tangent faces or very short edges produce unpredictable counts. Persistent homology (Section VIII) partially mitigates this.
2) χ **not sufficient**: use Betti numbers (Section VII) for stronger guarantees.
3) **Sheet metal restriction**: cell classification complete only for Definition I.1. Castings / injection-moulded parts need additional cell types.
4) **Geometric errors undetectable**: topological and geometric (Hausdorff) validation are complementary, not competing.
5) **Decomposition uniqueness unproven**: open problem (see MR-1b block).
6) l_p **ambiguity**: highly branched midsurfaces may produce ambiguous path counts; a combinatorial uniqueness proof is needed.

XII. Conclusion and Future Work

[FIX] FIX: Original Conclusion was 3 Sentences (62 words)

The original paper's conclusion was far too brief for a journal submission. The expanded conclusion below adds: summary of contributions, key findings, and a structured Future Work list. Target 200+ words.

This work proposes an algebraic topological framework for validating midsurface abstractions of thin-walled sheet metal solids. Two dual chain maps — dimension reduction (solid $\rightarrow$ midsurface) and dimension addition (midsurface $\rightarrow$ solid) — are derived via cellular decomposition and grounded in CW-complex theory. The Euler-Poincaré invariant validates entity counts in $O(|V|+|E|+|F|)$ time. Six shape cases confirm that predicted counts match actual midsurfaces and satisfy both Euler-Poincaré equations.

Key findings: (1) χ is necessary but not sufficient; Betti numbers provide stronger guarantees. (2) The transformation rules constitute a chain map (commutativity to be formally verified). (3) An error taxonomy classifies which errors each validation level detects.

Future work: formal chain-map commutativity proof; extension to variable-thickness parts; persistent-homology implementation; error injection experiments; integration with NN-based midsurface pipelines [32]; proof of decomposition independence.

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